

Dawn of High-Efficiency Solar Cells

To create highly efficient solar cells, researchers are looking at approaches like integrating PV systems with solar thermal components, light splitting, solar energy harvesting at night and diamond-based solar concentrators



RATHINDRA NATH BISWAS

The sun delivers 1.2×10^6 terawatts of power to the earth, which is about 10^4 times the rate at which human civilisation currently produces and uses energy. More energy from sunlight strikes in one hour than all the energy consumed by humans in an entire year. In fact, the solar energy resources dwarf all other renewable and fossil-based energy resources combined. Being green, it receives heightened attention as well.

A solar cell is basically a sandwich of semiconductor surrounded by layers of electrodes, which convert the sun's photons into electrons. Photovoltaic (PV) technology can directly convert

solar energy into electrical energy without any emission.

To provide a truly widespread primary energy source, solar energy must be captured, converted and stored in a cost-effective fashion, even if the device operates near the theoretical limit.

The theoretical efficiency limit for even an optimal single-bandgap solar conversion device is 31 per cent. It is because photons having energies lower than the absorption threshold of the active PV material are not absorbed, whereas photons having energies much higher than the bandgap release heat to the lattice of the solid. If all the photons were converted into electricity, the

theoretical efficiency limit would have been about 70 per cent.

PV modules shipped for commercial use now have a typical efficiency of 13-15 per cent. The losses are almost entirely due to small reflection, grid shading, heat loss, etc.

At present, the selling price of grid-connected PV electricity is about Rs 20 per kilowatt hour (kWh), whereas utility-scale electric power generation costs about Rs 2.50 per kWh. Hence for solar electricity to be cost-competitive with fossil-based electricity, there must be improvement in efficiency and manufacturing cost must be substantially reduced. If photons having energies greater than the bandgap of the absorbing material do not dissipate their excess energy as heat and instead produce more electrical energy, theoretical efficiency in excess of 60 per cent would be attainable.

Potential candidates for efficiency improvement

Direct and efficient use of solar energy is given high priority to address energy and environmental issues. The United Nations panel foresees big growth in renewable energy. In 2008 the world consumed around 492 exajoules (10^{18} joules) of energy and renewable energy made up 12.9 per cent of the total. The contribution of direct solar energy was only 0.1 per cent of the total energy consumed. It is estimated that by 2050 the direct solar energy could grow from 2008 level to about 100 exajoules per year. In the quest of green power, dedicated research work during the last decade has provided a new generation of renewable power source technologies at competitive price.